

TNA Core Language v1: The Structural Grammar of Axiomatic Necessity

Abstract

The Theory of Axiomatic Necessity (TNA) has been developed through a series of papers spanning mathematical logic, epistemology, philosophy of mind, artificial intelligence, ontology, and metaphysics. Although these works address different domains, they are unified by a single structural principle.

This document introduces the core language of TNA. It defines its primitive concepts, establishes the hierarchy of structural domains, and distinguishes formally proven results from their ontological consequences. In particular, we formulate the Structural Non-Derivability Corollary, which captures the impossibility of deriving instantiated realization from complete axiomatic description.

The purpose of this paper is not to introduce new applications but to provide the common grammar from which all previous and future TNA developments can be expressed.

1. Motivation

Most scientific theories begin with a formal language before they are applied to multiple domains. TNA has followed the opposite path. Its structural principles have been explored independently in mathematical logic, Gödel incompleteness, philosophy of mind, artificial intelligence, metaphysics, and structural ontology.

This document reconstructs those developments from a common foundation.

2. Primitive Domains

TNA distinguishes three structural domains:

- N_∞ : The domain of unlimited structural possibility. It is not a computable domain but

the source of all coherent structures that may become instantiated. It contains structural possibilities independently of any finite realization.

- N_0 : The domain of axiomatic description. A system belongs to N_0 whenever it is completely specified by formal rules, axioms, algorithms, or descriptive structures.

Examples include:

- Formal logical systems
- Mathematical theories
- Computer programs
- Physical descriptions
- Neural models

N_0 is the domain of representation.

- N_1 : The domain of instantiated realization. N_1 contains systems that are not merely described but executed, realized, or experienced. Examples include:

- Conscious experience
- Temporal execution
- Physical realization
- Embodied cognition
- Lived observation

N_1 is the domain of instantiation.

3. Primitive Structural Operations

The language of TNA uses five primitive operations:

- **Exteriority:** The sufficient condition for coherence cannot always be generated internally by the coherent system.
 - **Projection:** A coherent finite structure appears as a projection from a richer structural domain.
 - **Collapse:** Infinite structural possibility becomes a finite operational structure.
 - **Necessity:** Only structurally indispensable elements survive the collapse.
 - **Instantiation:** A description becomes an operational or experiential realization. Instantiation is not identical to description.
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4. Fundamental Theorem of TNA

The foundational theorem of TNA establishes that every coherent formal system possesses a minimal finite core of indispensable axioms. Consequently, the coherence of the system depends on structural elements that cannot be eliminated without loss of coherence.

This theorem is demonstrated using Compactness and monotonicity within first-order logic and establishes the metamathematical necessity of indispensable axioms. This theorem constitutes the formal foundation of TNA.

5. Structural Non-Derivability Corollary

The preceding theorem concerns the existence of indispensable axioms within formal systems. TNA extends its structural interpretation by distinguishing descriptive closure from instantiated realization.

Let:

- N_0 denote a complete axiomatic description of a system.
- N_1 denote the instantiated realization of that system.

Then:

$$\boxed{N_0 \not\vdash N_1}$$

Structural Non-Derivability Corollary. *A complete axiomatic description does not, by itself, entail the instantiated realization of the system it describes.*

The transition from description to realization is therefore a structural boundary rather than a merely informational gap. This statement is presented as a corollary of the structural interpretation of the Theorem of Axiomatic Necessity. It is not the original metamathematical theorem itself but an ontological consequence obtained after introducing the distinction between descriptive and instantiated domains.

6. Structural Interpretation

The distinction between N_0 and N_1 provides a common language for interpreting problems that previously appeared unrelated. Examples include:

- Gödel incompleteness

- The Knowledge Argument (Mary)
- Artificial consciousness
- AI alignment
- Structural ontology
- The relation between physics and metaphysics

Each becomes an instance of the same invariant:

$$N_0 \not\sim N_1$$

7. The TNA Grammar

The complete structural grammar may be summarized as follows:

$$N_\infty \longrightarrow N_0 \not\sim N_1$$

where:

- N_∞ provides structural possibility,
- Projection generates finite axiomatic structures,
- Collapse selects minimal coherent cores,
- N_0 describes,
- N_1 realizes.

Every subsequent application of TNA is obtained by assigning concrete interpretations to these symbols while preserving the same structural grammar.

8. Conclusion

TNA is not a theory about consciousness, artificial intelligence, mathematics, or metaphysics individually. It is a theory about the structural relation between possibility, description, necessity, and realization.

The Theorem of Axiomatic Necessity establishes the formal existence of indispensable external structure. The Structural Non-Derivability Corollary extends this result into a general principle governing the relation between complete description and instantiated realization. Together they define the core language from which the entire TNA framework can be systematically

developed.

References

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2. Bresciano, C. (2025). The Theorem Nobody Wants: Metamathematical Necessity of Axioms via Reduction to Incoherence v1.3. Zenodo. DOI : 10.5281/zenodo.19192655